

DRUGS USED IN THE TREATMENT OF ANGINA PECTORIS

ASSOC. Prof. Dr. Wanida Sukketsiri
Division of Health and Applied Sciences,
Faculty of Science

wanida.su@psu.ac.th

Pr.201

วัตถุประสงค์การเรียนรู้

- อธิบายแนวทางในการรักษาโรคกล้ามเนื้อหัวใจขาดเลือด
- บอกชื่อยาหลักและกลุ่มยาที่ใช้รักษาโรคกล้ามเนื้อหัวใจขาดเลือด
- อธิบายกลไกการออกฤทธิ์และเภสัชวิทยาของยาที่ใช้รักษาโรคกล้ามเนื้อหัวใจขาดเลือด
- เลือกใช้ยารักษาโรคกล้ามเนื้อหัวใจขาดเลือดได้อย่างเหมาะสม

ANGINA PECTORIS

Symptom of ischemic heart disease

Insufficient blood supply relative to
oxygen requirement



Anoxia in some parts of the heart

ANGINAL PAIN

*Sudden,
severe substernal pain*

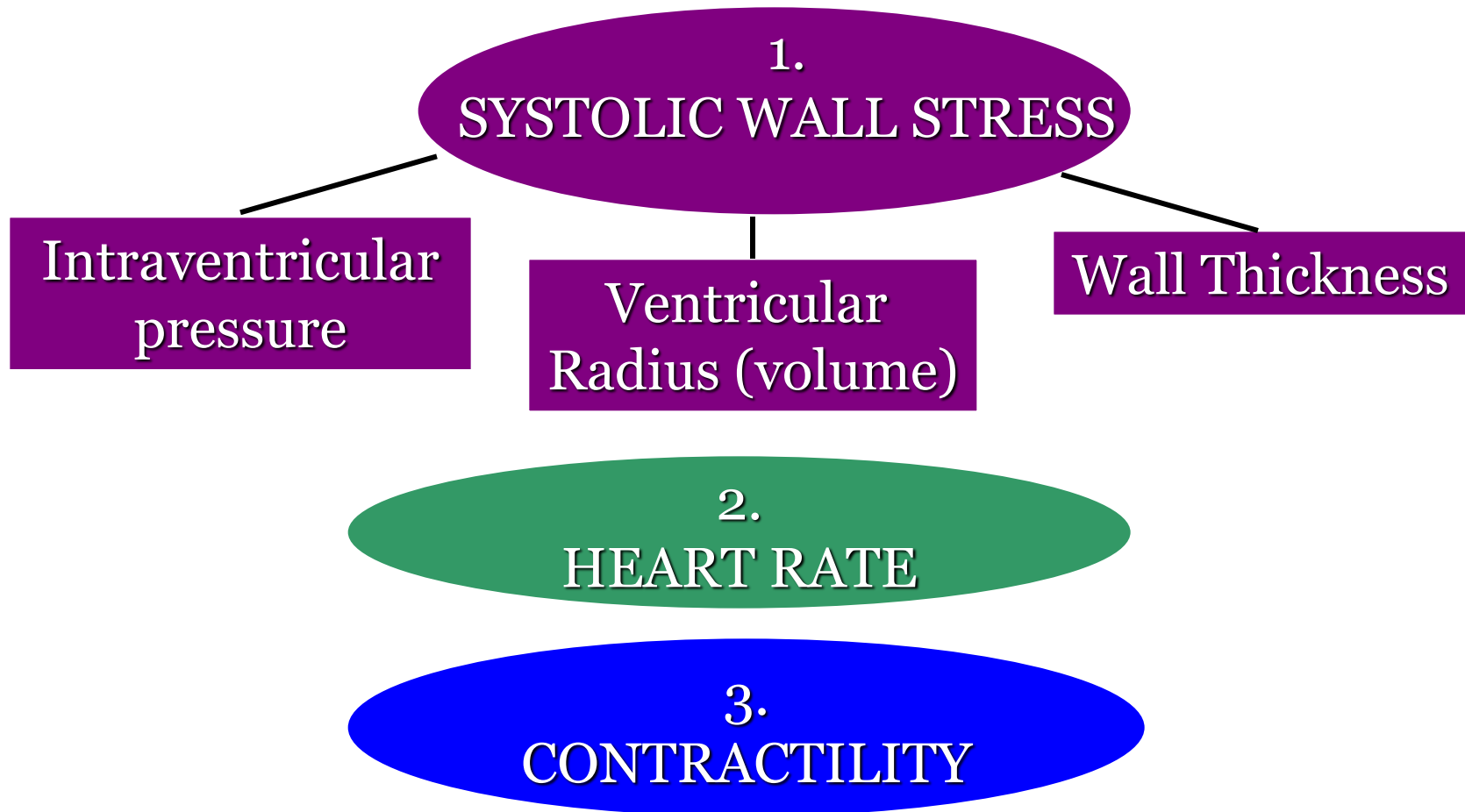
**MYOCARDIAL
INFARCTION**

CHF

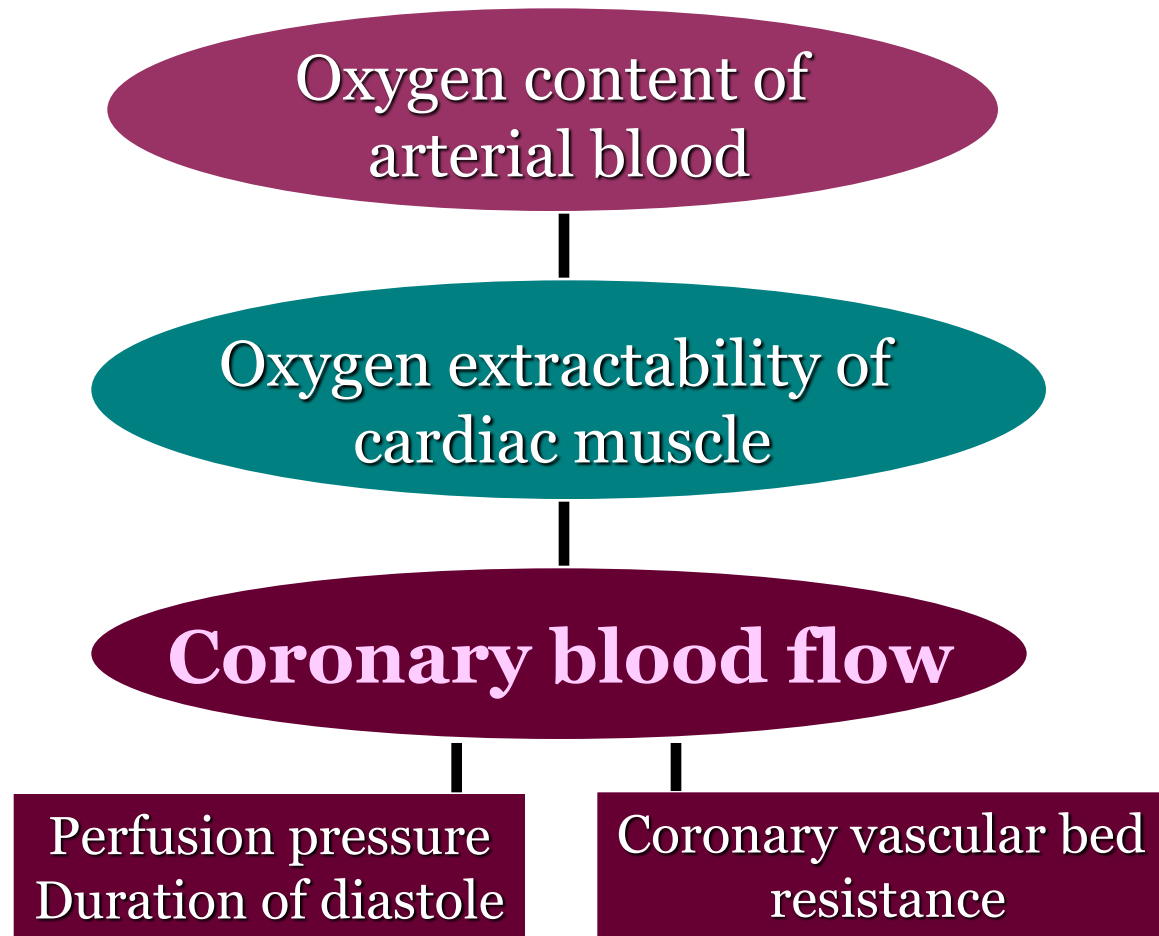
ARRHYTHMIAS

DEATH

DETERMINANTS OF MYOCARDIAL OXYGEN DEMAND



DETERMINANTS OF MYOCARDIAL OXYGEN SUPPLY



FACTORS DECREASING OXYGEN SUPPLY

Decreased Oxygen Transport

- ❖ *Anemia****
- ❖ *Chronic lung disease*
- ❖ *Cigarette smoking*

Increase resistance to flow

- ❖ *Atherosclerosis ****
- ❖ *Aortic stenosis*

FACTORS INCREASING OXYGEN DEMAND

- ❖ *Congestive heart failure*
- ❖ *Increase intracardiac pressure****
- ❖ *Increase heart rate: Thyrotoxicosis ****
exercise, drugs

❖ **STABLE ANGINA** (*exertional, typical, classical, atherosclerotic*)

- ** Severe and extensive organic stenosis of coronary artery*
- ** Exercise, cold, emotion, stress, eating-precipitate episode*
- ** Underlying pathology is usually **atherosclerosis***

❖ **VASOSPASTIC ANGINA** (*varaint, Prinzmetal's*)

- ** Transient vasospasm of the coronary vessels*
- ** Usually associated with underlying atheroma*
- ** Chest pain develop at rest*

❖ **UNSTABLE ANGINA** (*preinfarction, crescendo angina, angina at rest*)

- ** **Atherosclerotic plaque rupture**, endothelial injury, ulceration at site of coronary, platelet activation, coronary artery thrombus formation*
- ** Coronary spasm, artery stenosis*
- ** Associate with a change in character, frequency and duration of angina in patient with stable angina*

TREATMENT OF ANGINA PECTORIS

- Alleviate symptoms
- Control & modify risk factors
 - Smoking, HT, DM, obesity, dyslipidemia, stress
- $\uparrow O_2$ supply and $\downarrow O_2$ demand
- Pharmacologic therapy
 - Nitrate, beta-blockers, CCBs, newer anti-ischaemic drugs (ivabradine, nicorandil, ranolazine, trimetazidine)
- Intervention therapy
 - Percutaneous coronary angioplasty (PTCA)
 - Coronary artery bypass grafting (CABG)

DRUGS USED TO TREAT ANGINA

Drug groups currently approved for use in Angina – *All decreases myocardial oxygen requirement of the heart*

- ❖ CALCIUM CHANNEL BLOCKERS
- ❖ β -ADRENOCEPTOR ANTAGONISTS
- ❖ ORGANIC NITRATES
- ❖ Newer anti-ischaemic drugs (ivabradine, nicorandil, ranolazine, trimetazidine)

Calcium Channel Blocking Drugs

CALCIUM ANTAGONISTS

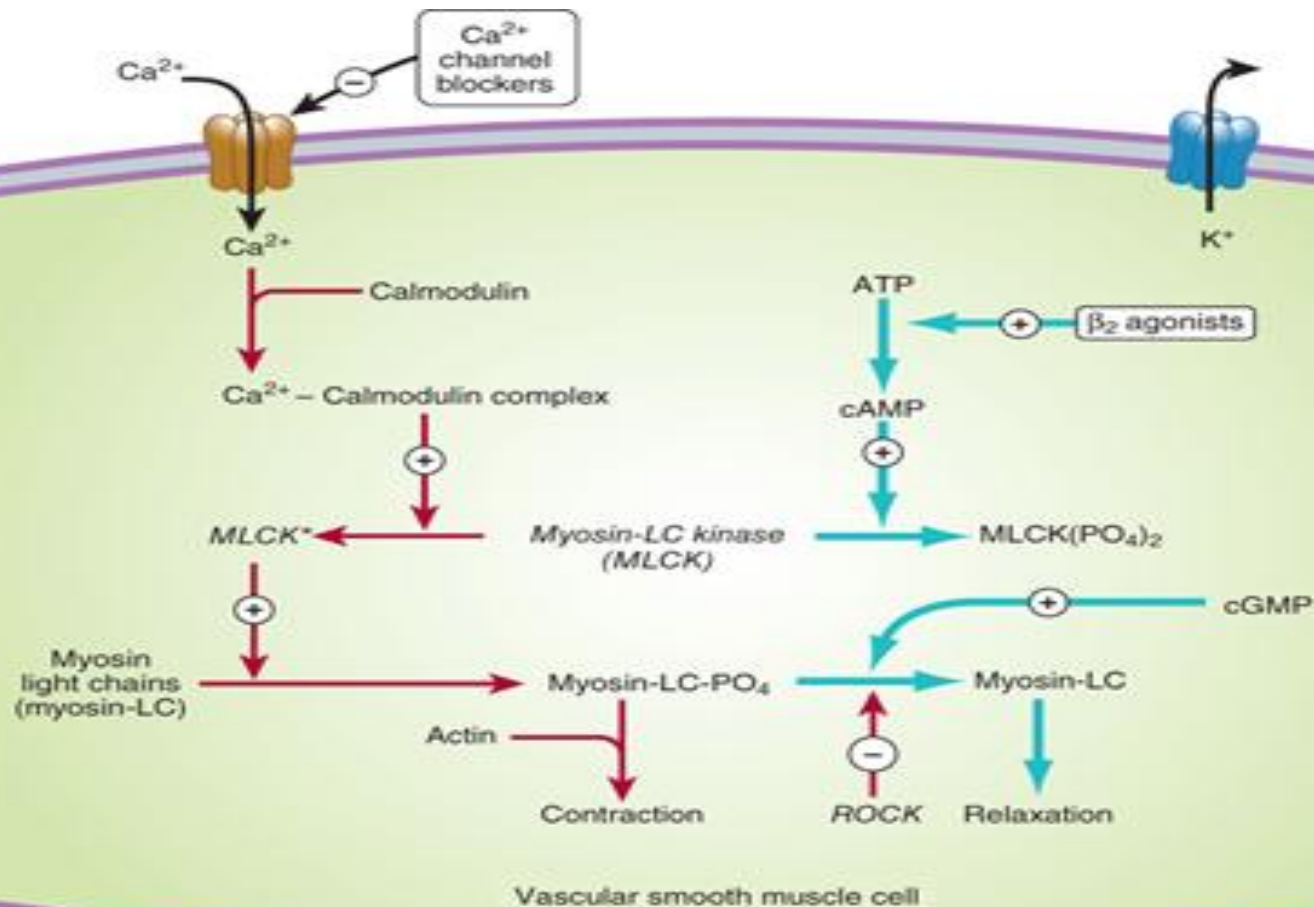
DIHYDROPYRIDINES (DHP):

- ❖ *Amlodipine, Nifedipine, Nicardipine etc*
- ❖ *bind with α_1 -subunit (N- sites) of L-type Ca^{2+} channel*
- ❖ *vascular smooth muscle > myocardium*
- ❖ *peripheral or coronary vasodilatation*

NONDIHYDROPYRIDINES (non-DHP):

- ❖ *Verapamil, Diltiazem*
- ❖ *bind with α_1 -subunit (N- sites) of L-type Ca^{2+} channel*
- ❖ *myocardium > vascular smooth muscle*
- ❖ *decrease HR and contractility*

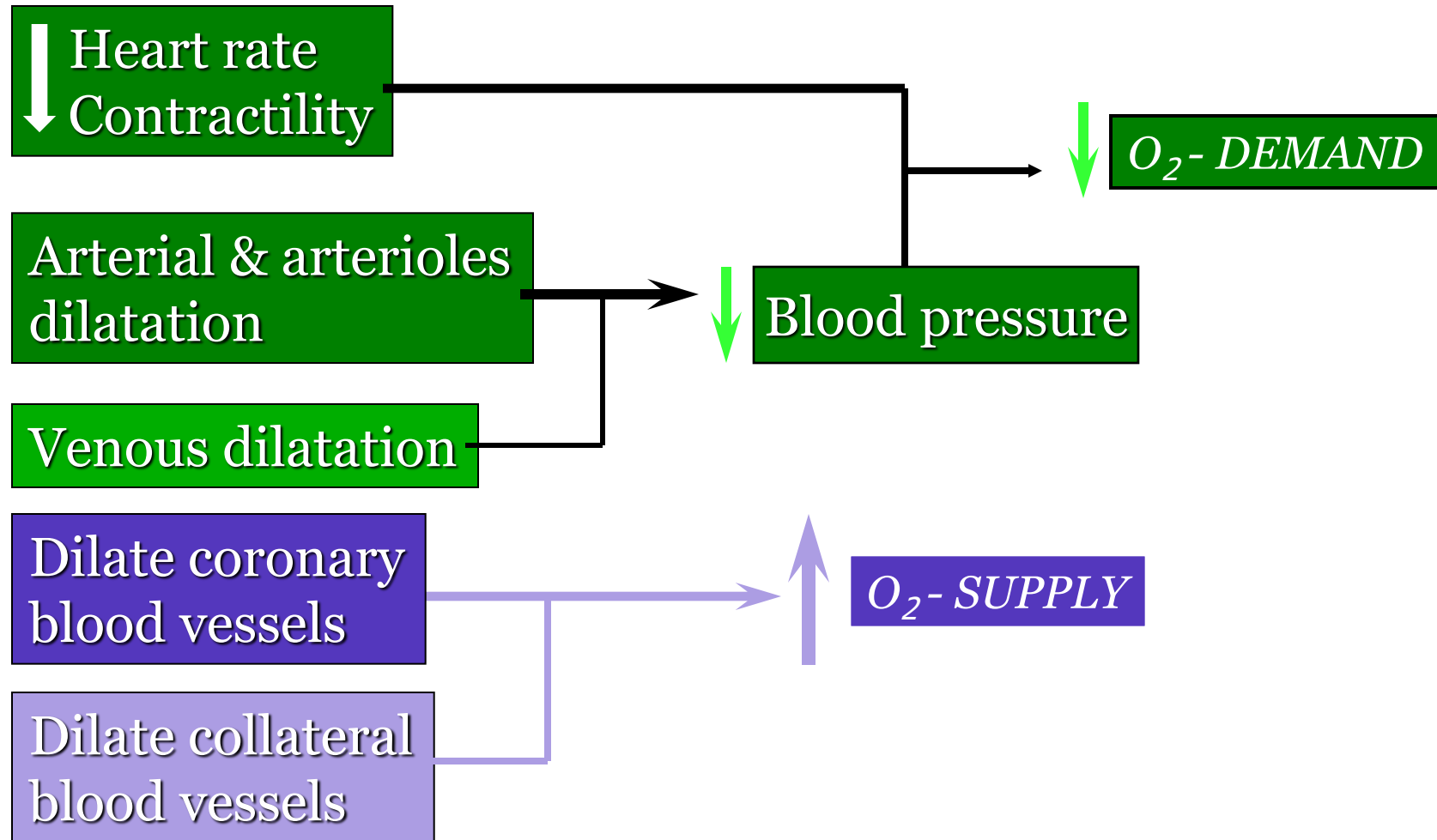
CCBs – Mechanisms of Action



Hemodynamic Effects of CCBs

Effect	Verapamil	Diltiazem	Nifedipine
Peripheral vasodilatation	↑	↑	↑↑
Coronary vasodilatation	↑↑	↑↑↑	↑↑↑
Preload	0	0	0/
Afterload	↓↓	↓↓	↓↓↓
Contractility	↓↓	0/↓	↑/↓
Heart rate	0/↓	↓	↑/0
AV conduction	↓↓	↓	0

CALCIUM ANTAGONISTS



Comparative Adverse Effects

		Diltiazem		Verapamil		Dihydropyridines	
Overall		0-3%		10-14%		9-39%	
Hypotension		++		++		+++	
Headaches		0		+		+++	
Peripheral Edema		++		++		+++	
Constipation		0		++		0	
CHF (Worsen)		0		+		0	
AV block		+		++		0	
Caution w/beta blockers		+		++		0	

CCBs - Monitoring

- Heart rate
- Blood pressure
- Anginal symptoms
- Signs of CHF
- Adverse effects

Contraindications for CCBs

Contraindication	Verapamil	Nifedipine	Diltiazem
Hypotension	+	++	+
Sinus bradycardia	+	0	+
AV conduction defects	++	0	++
Severe cardiac failure	++	+	+

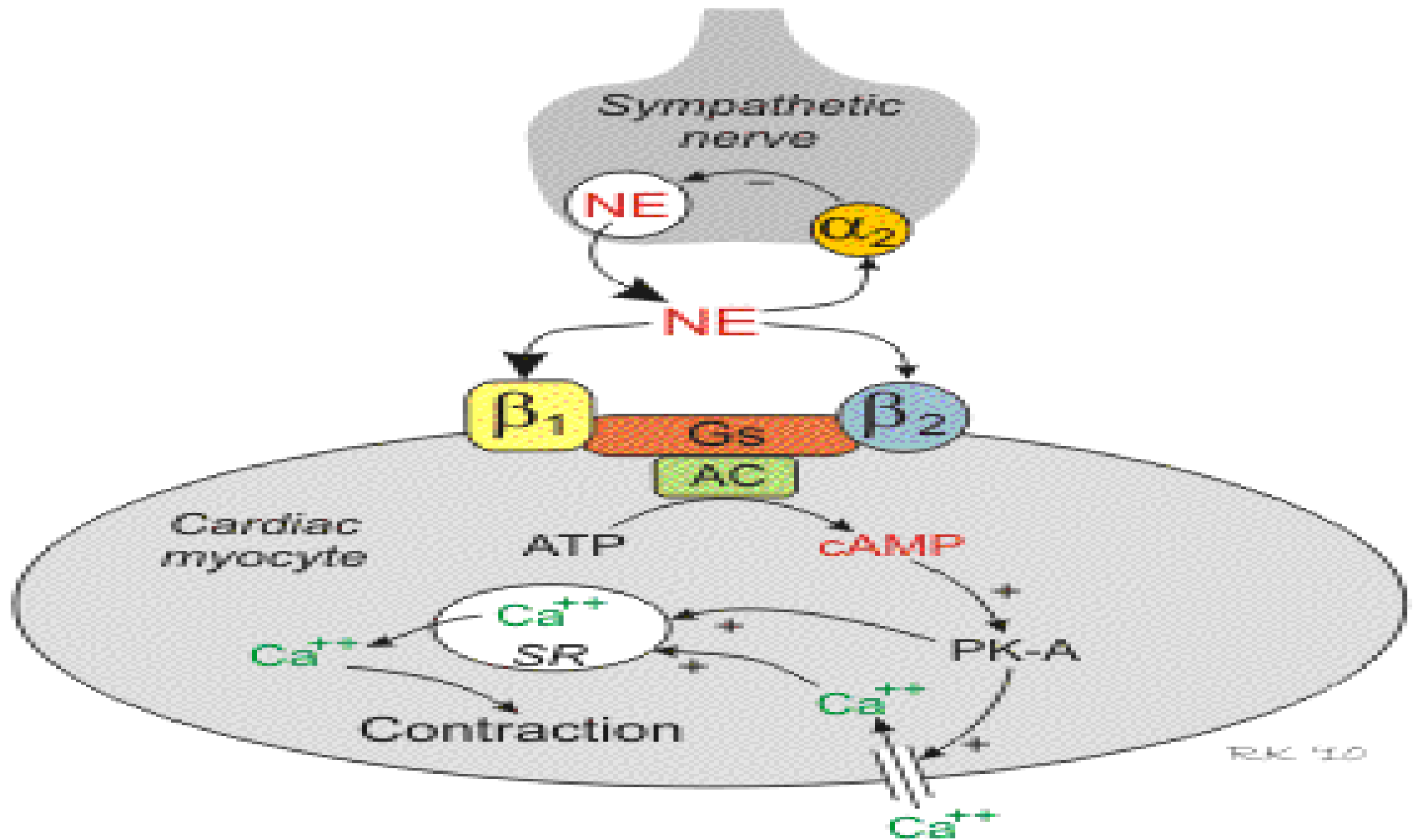
Widespread use of CCBs

- Angina pectoris
 - Variant angina : coronary blood flow rather than peripheral
 - Exertion angina : effective in treatment
 - Unstable angina : may offer an additional approach to treatment
 - MI : no evidence
- Hypertension
- Treatment of supraventricular arrhythmias: Atrial Flutter, Atrial Fibrillation, Paroxysmal SVT

β -ADRENOCEPTOR ANTAGONISTS

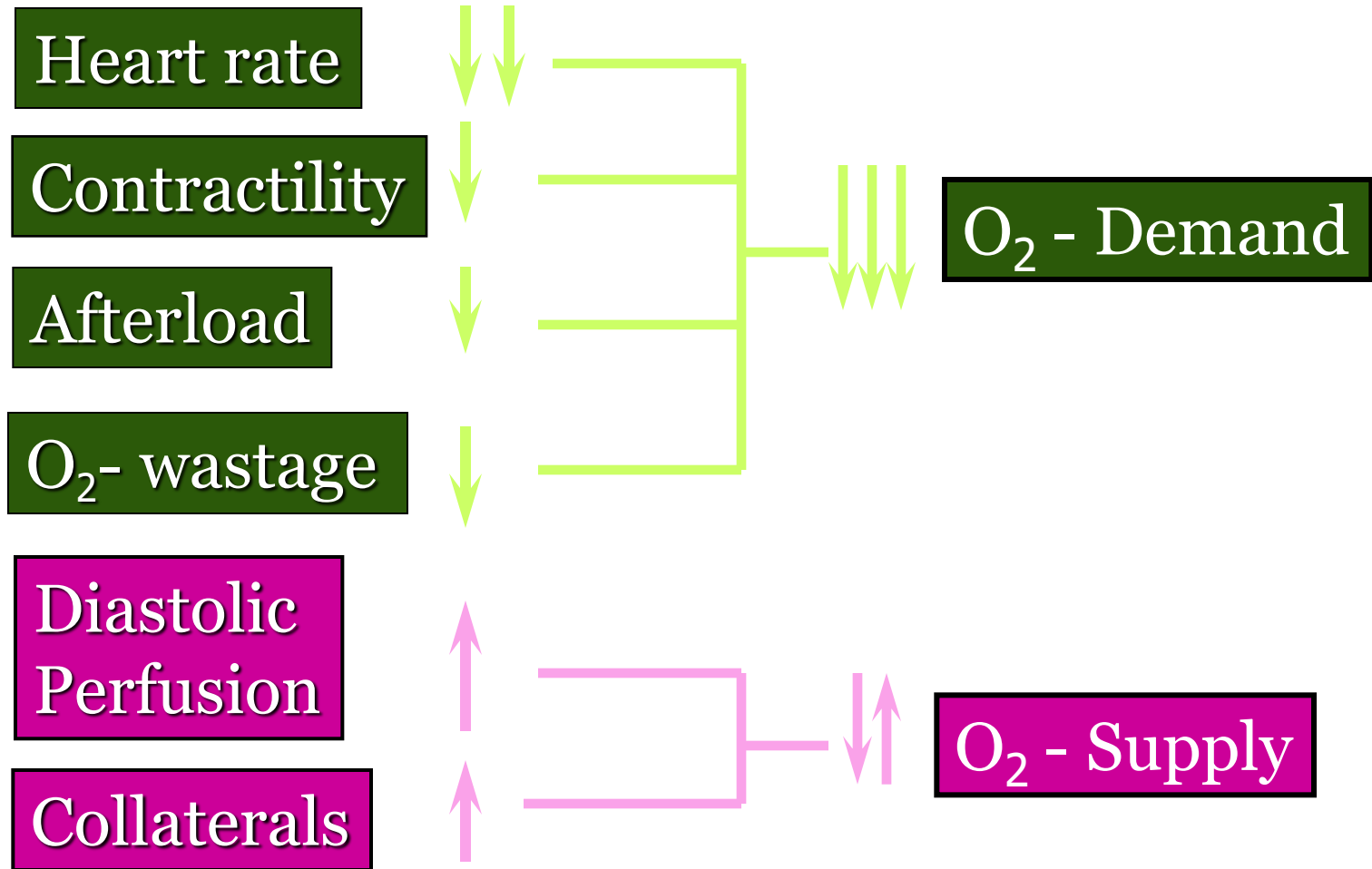
- Selective β_1 -blockers: metoprolol, atenolol
- Non-selective β_1 -blockers: propranolol
- Labetalol : β -blockers with α -blockers
- Nebivolol : β -blockers with vasodilation (NO pathway)
- Acebutolol : β -blockers with intrinsic sympathomimetic activity

β -ADRENOCEPTOR ANTAGONISTS



Abbreviations: NE, norepinephrine; Gs, G-stimulatory protein; AC, adenylyl cyclase; PK-A, cAMP-dependent protein kinase; SR, sarcoplasmic reticulum

β -ADRENOCEPTOR ANTAGONISTS



β-ADRENOCEPTOR ANTAGONISTS

Adverse drug reaction

- Bradycardia
- Bronchospasm, wheezing, cold extremities
- Hyperglycemia
 - Diabetic: hypoglycemia
- Heart failure: edema, dyspnea
- Other...fatigue, dyslipidemia

Contraindication

- Asthma, DM, SA, AV nodal dysfunction

β -ADRENOCEPTOR ANTAGONISTS

Therapeutic use

- Exertion angina : effective in treatment
- Unstable angina: effective in reducing recurrent of ischemia and the risk of progression to acute MI
- MI: decreased mortality in MI
- Other uses: HT, arrhythmia, HF

ORGANIC NITRATES

NITRATES & NITRITES

Chemistry:

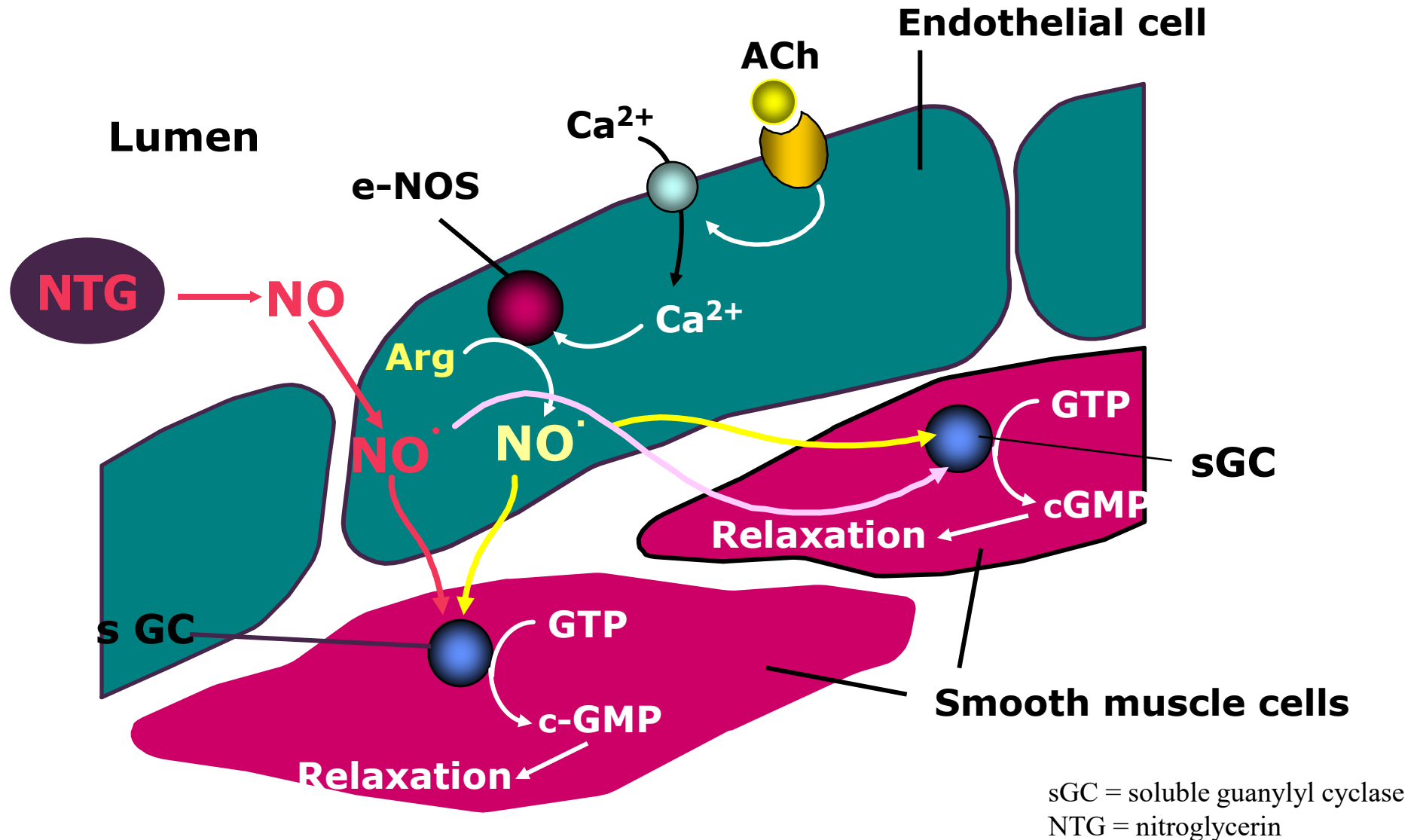
nitric and nitrous acid esters of polyalcohol

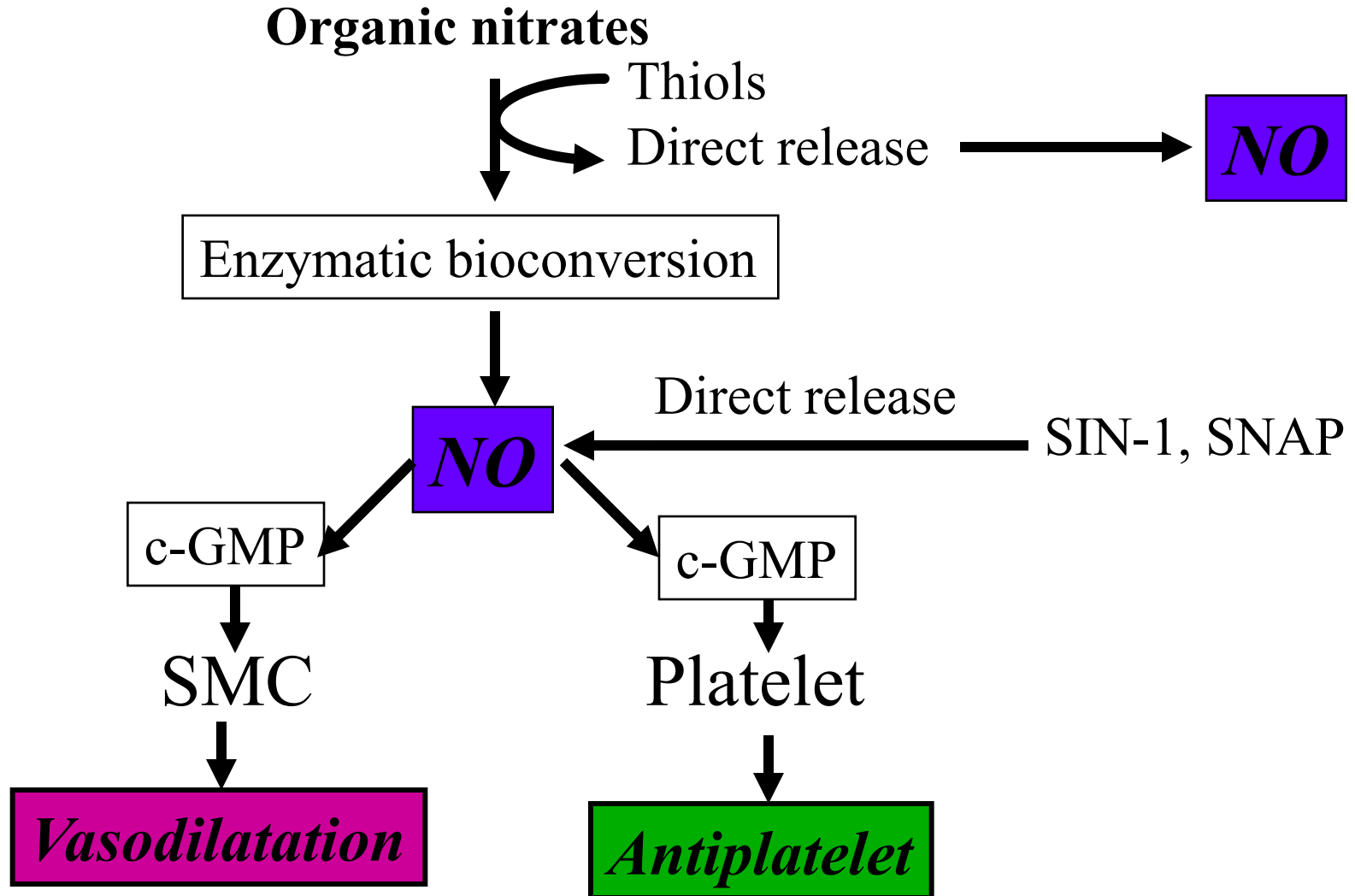
Glyceryl trinitrate, Isosorbide dinitrate, Isosorbide mononitrate etc

Pharmacokinetics:

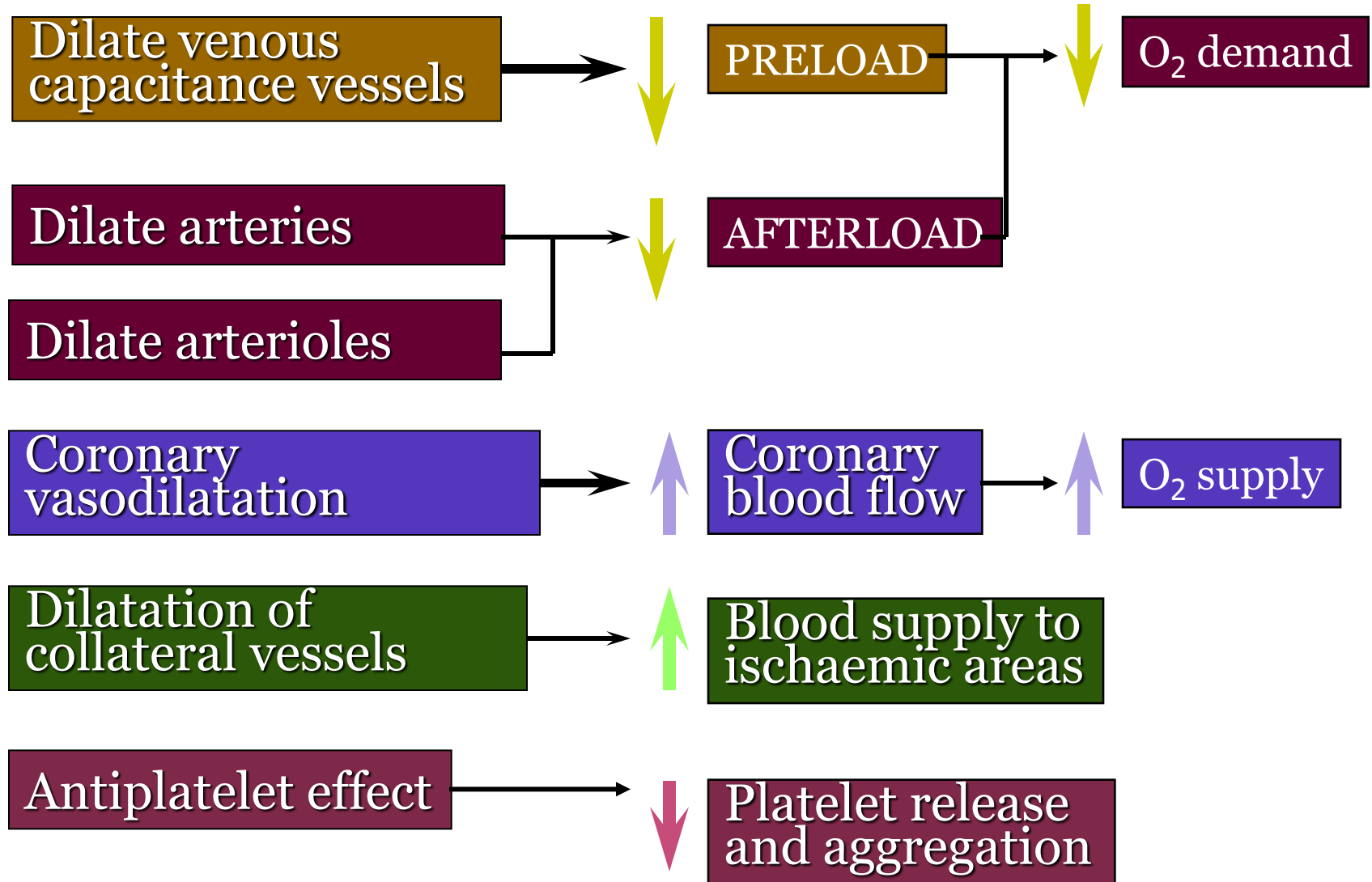
- ❖ *liver **nitrate reductase** destroys drug rapidly*
oral bioavailability <10-20% (high hepatic first pass metabolism)
- ❖ *absorbed unchanged nitrate - $T/2 = 2-8$ min*
- ❖ *nitroglycerin metabolites:*
 - *two mononitro forms*
 - *two dinitroglycerins- sig. vasodilator efficacy*
- ❖ *excretion - **glucuronide** derivatives of the denitrated metabolites - via **kidney***

Signal transduction pathway involving NO and c-GMP that leads to vasodilation





ORGANIC NITRATES



ORGANIC NITRATES

preparations available

- ❖ Sublingual NTG tablets (*0.3-0.6 mg tab*) ***
- ❖ NTG spray (*0.4 mg/ metered dose aerosol*)
- ❖ Oral NTG (*sustained-release: 2.6, 6.5, 9 mg tab*
2.5, 6.5, 9, 13 mg cap)
- ❖ NTG buccal tablets (*2, 3 mg*)
- ❖ NTG topical ointment (*20 mg/ml*)
- ❖ NTG transdermal patches (*0.1-0.8 mg/h*)
- ❖ Intravenous NTG (*0.5, 5 mg/ ml*)

ORGANIC NITRATES

precautions & side effects

PRECAUTIONS:

keep in airtight container (*volatile & explosive*)

SIDE EFFECTS:

Common – *Severe throbbing headaches*

Dizziness, Flushing

Serious - *Syncope & hypotension*

tachycardia, methemoglobinemia

DRUG INTERACTION

- ❖ Sildenafil, vardenafil, tadalafil used in erectile dysfunction
- ❖ Risk of hypotension

NEW ANTIANGINAL AGENTS

Drugs	Mechanism of action	Dose	Contraindications	Major adverse effects
Ivabradine	- Reduction of pacemaker activity in SA node - Lower heart rate and thus more blood to flow to the myocardium	Oral dose of 5 to 7.5 mg twice daily; 2.5 mg twice daily (>75 years of age or according to symptoms and heart rate)	Severe liver disease, low heart rate or heart rhythm disorder, allergy	Bradyarrhythmia, visual disturbances
Nicorandil	- Potassium channel opener with nitrovasodilator (NO donor) actions	10 mg twice daily or 5 mg twice a day (if headache); maximum dose of 20 mg twice daily	Low BP, hypovolemia, severe HF, concomitant use of phosphodiesterase 5 inhibitors and/or soluble guanylate cyclase stimulators	Gastrointestinal ulcerations, skin and mucosal ulcerations

NEW ANTIANGINAL AGENTS

Drugs	Mechanism of action	Dose	Contraindications	Major adverse effects
Ranolazine	- Inhibition of the late inward sodium channel - Reduces intracellular sodium and, subsequently, intracellular calcium overload	375 mg twice daily to 750 mg twice daily (prolonged-release tablets)	Severe renal insufficiency, moderate or severe hepatic impairment; concomitant administration of potent CYP3A4 inhibitors or class Ia or class III anti-arrhythmic drugs (such as dofetilide, sotalol)	Nausea, constipation, dizziness, headache, asthenia, prolongation of the QT interval
Trimetazidine	- Inhibition of the reduction of adenosine triphosphate, stimulation of glucose consumption - Inhibit mitochondrial 3-ketoacyl coenzyme A thiolase	20 mg three times a day or 35 mg twice daily	Parkinson's disease, tremors and movement disorders, severe renal impairment	Parkinsonism symptoms, gastric discomfort, headache, movement disorders

	Standard therapy	ชีพจรเต้นเร็ว (เช่น >80 bpm)	ชีพจรเต้นช้า (เช่น <50 bpm)	มี LV dysfunction หรือ heart failure	ความดันโลหิตต่ำ
ขั้นที่ 1	BB หรือ CCB	BB หรือ non-DHP-CCB	DHP-CCB	BB	Low-dose BB หรือ low-dose non-DHP-CCB
	↓	↓	↓	↓	↓
ขั้นที่ 2	BB + DHP-CCB	BB + CCB	เปลี่ยนเป็น LAN	เพิ่ม LAN หรือ ivabradine	เปลี่ยนเป็น ivabradine, ranolazine หรือ trimetazidine
	↓	↓	↓	↓	↓
ขั้นที่ 3	เพิ่ม 2 nd line drug	BB + ivabradine	DHP-CCB + LAN	เพิ่ม 2 nd line drug ตัวอื่น	ใช้ 2 nd line drug สองตัวร่วมกัน
			↓		
ขั้นที่ 4			เพิ่ม ranolazine หรือ trimetazidine		

ภาพที่ 3 แนวทางการใช้ยาลดอาการเจ็บหน้าอกจากกล้ามเนื้อหัวใจขาดเลือด

Other drugs

- **Antiplatelet** – prevent cardiovascular events

Aspirin, Clopidogrel, Ticagrelor, Prasugrel

- **Anticoagulant**

- Non-vitamin K antagonist oral anticoagulant (NOACs)
or vitamin K antagonist (VKA)
- LMWH (s.c.) or thrombolytic agents

- **Lipid lowering drugs**